

3.6 Mathematical Requirements

In order to develop their skills, knowledge and understanding in science, candidates need to have been taught, and to have acquired competence in,

the appropriate areas of mathematics relevant to the subject as set out below.

	Candidates should be able to:
Arithmetic and numerical computation	• recognise and use expressions in decimal and standard form • use ratios, fractions and percentages • make estimates of the results of calculations (without using a calculator)
Handling data	• use an appropriate number of significant figures • find arithmetic means • construct and interpret frequency tables and diagrams, bar charts and histograms • have sufficient understanding of probability, eg to understand how genetic ratios arise • understand the principles of sampling as applied to scientific data • understand the importance of chance when interpreting data • understand the terms mean, median and mode • use a scatter diagram to identify a correlation between two variables • use a simple statistical test • make order of magnitude calculations
Algebra	• understand and use the symbols: $=$, $<$, $<<$, $>>$, $>$, ∞, $\sim$ • substitute numerical values into algebraic equations using appropriate units for physical quantities
Graphs	• translate information between graphical, numerical and algebraic forms • plot two variables from experimental or other data.